

LTA Nebula X Hackathon - PS3

By: Rail Trains-parency

[1] Background Research

Prior to getting hold of the data and key information, the Team has been researching the Problem Statement context publicly available: Every train is already telling us what it needs.. Research on the target users of technicians and operations control staff surfaced the same two problems wearing different clothes: too much fragmented data, and none of it presented in a way that turns into an actual decision. With these pain points in mind, we built around timing, not just detection — a tri-colour scale instead of raw numbers, dashboards filtered to what actually matters to each user, reports that work at both the executive and technical level, and a loop that closes when a fix in the field feeds back into the system instead of vanishing into a log.

[2] Model Building - Team's Approach

Prior to the building of the app, the Team has split the subsystems equally for the model development of each subsystem. This was done due to the vastly differing dataset types provided, hence different types of input features, models and output features have to be taken into consideration and focus.

The in-depth analyses of each subsystem are as below:

[i] ACV

This subsystem locates which car within an 8-car MRT has a refrigerant leak and is struggling most to reach its cooling setpoint, detected by tracing the cabin-temperature-minus-target gap while actively cooling — a value that deviates from the other cars on the same train creates a trace for identifying the faulty car.

Eight features per car were computed from each car's own telemetry and robust-z-scored within-file (e.g. cooling-gap mean/p90/p99 while in cooling mode), to ensure comparability across files with differing ambient regimes and sensor scales.

Given only 6 labelled cases, six unsupervised anomaly-scoring baselines (z-score, Mahalanobis, Isolation Forest, LOF, PCA reconstruction, ensemble) were compared against a supervised logistic-regression layer via nested leave-one-file-out cross-validation (LOFO-CV). Mahalanobis and logistic regression tied at 0.979; Mahalanobis was chosen since it needs no training data and adapts to new file schemas without retraining. The subsystem achieved a competition rank-decay score of 0.979 under nested LOFO-CV — a deliberate metric choice, not a stand-in for F1 or accuracy: since the deliverable is "which car should the maintenance team

inspect first," a prediction off by one rank (2nd instead of 1st) is still operationally useful and earns partial credit, whereas a fixed-threshold metric would score a near-miss identically to a complete failure. This figure warrants one caveat: 5 of the 6 files contain an obvious cooling-gap signal any method would catch, leaving `acv_case_04.xlsx` (richer schema) as the only real stress test — its faulty car ranks 2nd, not 1st, once leaky pressure/compressor features are removed, the honest cost of a leakage-audited feature set.

Three design decisions filled gaps the brief left underspecified: "actively cooling" was operationalized from each file's own status/mode columns rather than a fixed schedule, since cooling behaviour varies with ambient load; car identity and count were read from each file's own headers rather than hardcoded to 8 cars, consistent with handling both the standard and richer schemas; and LOFO-CV was adopted as the natural split given only 6 labelled files and per-file scoring, since no other grouping axis exists.

[ii] Door

`Train.csv/Test.csv` contain rows only during door motion, so timestamps jump from ~20ms apart (mid-cycle) to several seconds apart (between cycles). Segmentation uses a gap-detection rule — split wherever the timestamp delta exceeds 2.0s — that has been verified against `Train_Segments_Answer.csv`: 110/110 segments match exactly before being trusted on the unlabelled `Test.csv` stream.

There are approximately 121 candidate features per cycle. These features include summary statistics over motor current, voltage, back-EMF, and door-leaf position; a $\text{current} \times \text{voltage}$ energy proxy; travel-path length, displacement, overshoot ratio, and direction reversals; command/flag transition counts and on-fractions. Features with near-constant and highly-collinearity are dropped unsupervised before modelling, reducing the number of kept features to 33.

Inside every fold, the pipeline scales, SMOTE-balances the training fold only, applies SelectKBest, and fits an L1-penalised logistic regression — SMOTE is never applied to the held-out fold, which stays representative of the real class distribution. Seven candidates were compared under repeated stratified 5-fold CV (10 repeats, 50 splits): Random Forest scored marginally higher ($F1[\text{Abnormal}] = 1.000 \pm 0.000$) than L1 logistic regression (0.998 ± 0.013), ahead of L2 logistic regression, XGBoost, Gradient Boosting, Decision Tree, and a small MLP. L1 logistic regression was selected over the higher-scoring Random Forest for a reason the cross-validation numbers alone don't surface: on the actual test set, Random Forest and XGBoost flag two cycles as Abnormal that every logistic-regression- and gradient-boosting-family model calls Normal. Random Forest's own predicted probability for one disputed cycle is $P(\text{Abnormal}) = 0.519$ — genuinely uncertain, despite a "perfect" CV score — suggesting the 1.000 reflects how separable this particular 110-cycle training set happens to be, rather than robustness on unseen data. The

reported metric is F1 on the Abnormal class (the rarer, business-relevant class), with macro-F1 and ROC-AUC as supporting context: $F1[\text{Abnormal}] = 0.998 \pm 0.013$ and $\text{ROC-AUC} = 1.000$ across 50 validation splits. The competition's own IoU-weighted F1 scoring, which judges both segment timing overlap and label, is applied identically at submission time.

This evaluation strategy follows directly from a constraint the brief leaves open. Train.csv is a single continuous stream from one door, not multiple files or labelled operating conditions, so there is no natural axis along which to split the data. With only 110 labelled cycles (30 Abnormal), a single fixed train/validation split would leave too few validation positives for a stable estimate — a single flipped prediction can swing F1 by roughly 0.15–0.3. Every cycle is therefore used as both training and validation data, via repeated stratified 5-fold CV $\times$ 10 repeats (50 train/validation splits), each fold preserving the $\sim 27\%$ Abnormal ratio, with the result reported as a distribution (mean $\pm$ std) rather than a single noisy point estimate.

[iii] Rail Corrugation

Each file is a 1-second, 10kHz recording: a raw speed-sensor toggle signal with vibration and shock pairs for 64 axle-box positions (8 cars $\times$ 8 positions). For each car, the odd-numbered positions belong to Side I, while even-numbered positions belong to Side II, judged independently from the same file.

Car and position are parsed directly from each column's header text via regex, rather than assumed from a fixed column order, so a reordered or partial column set can't silently misassign a channel. All 128 signal channels are processed independently rather than aggregated across cars, since a corrugation fault is localised and aggregating risks averaging it away. For each channel (plus the rotating-speed column), 9 time- and frequency-domain statistics are extracted — 1,161 raw features per file, narrowed by SelectKBest (ANOVA F-test, refit per fold) to ~ 50 class-discriminative features. A near-constant rotating-speed signal (present on 38 of 272 files) makes kurtosis undefined; these are filled with 0.

With 234 Normal / 24 Side II / 14 Side I files out of 272 — Side I comprising just 5.1% of the data — class imbalance was handled through weighting mechanisms rather than resampling: `class_weight='balanced'` (or `'balanced_subsample'` for Random Forest) on every candidate that supports it; `sample_weight` computed via `compute_sample_weight`, fresh on each fold's training portion for XGBoost, which doesn't support `class_weight` for 3 classes; StratifiedKFold throughout so no fold is starved of the rare classes; loosened `min_samples_leaf/min_samples_split` (3/5) so a tree can still isolate a leaf from Side I's 14 examples; and macro F1 as the model-selection metric itself, since a model that never predicts Side I could otherwise exceed 90% accuracy while scoring 0 on that class. Six candidates — LASSO (L1) and Ridge (L2) logistic regression, decision tree, random forest, XGBoost, and SVM with an RBF kernel — were compared under identical nested CV

at three seeds (42, 7, 123). SVM won on both criteria that matter here: highest mean macro F1 (0.621 ± 0.050) and the lowest run-to-run variance across seeds, ahead of LASSO (0.598), Ridge (0.591), XGBoost (0.579), Random Forest (0.571), and Decision Tree (0.482). The reported metric is macro F1 across the three classes, matching the organisers' scoring exactly (Info Kit Section 4) — 0.621 ± 0.050 across the 3-seed nested-CV sweep — deliberately not plain accuracy, so the model is credited for detecting the rare Side I/Side II cases and not just the common Normal case.

This validation design follows from a gap the brief leaves open. No file-level grouping exists to split by, since each file is already one independent unit, so nested cross-validation (5-fold outer / 3-fold inner) was adopted so that hyperparameter tuning never sees the fold it is later scored on. This was repeated at three seeds (42, 7, 123) because, with only 14 Side I examples in total, which files land in which fold measurably changes the ranking between candidate models — a single-seed comparison isn't a trustworthy basis for a final choice.

[iv] SMH

Each training file is a one-column stress time series. Rainflow cycle counting converts the raw signal into discrete stress cycles, from which physically-motivated damage-proxy features are built, rather than feeding the raw time series into a model directly.

Time-domain features (mean stress, RMS, crest factor, kurtosis, mean-crossing rate) plus rainflow-derived features: pseudo-damage sums ($\sum \text{count} \times \text{amplitude}^{3/5}$, log-transformed), a Stress-Weighted-Temperature (SWT) diagnostic ratio, and a 24-bin load-spectrum histogram with global bin edges so `cycle_bin_X` means the same thing across every file. An `is_flat_signal` flag and `n_samples` feature prevent a flat signal or varying recording length from confounding the damage estimate.

Seven candidates — Ridge, ElasticNet, KNN+PCA, SVR, Random Forest, HistGradientBoosting, and XGBoost — were compared under nested CV, using a RandomizedSearchCV inner loop and a RepeatedKFold outer loop (5 folds $\times$ 3 repeats, 15 outer evaluations). A VIF-based feature selector runs inside every fold, protecting compositional histogram-bin features and any monotonic-constrained feature from removal, since a monotonic model's constraint set must stay fixed across folds; the gradient-boosting candidates additionally receive monotonic constraints — damage cannot decrease as RMS, peak-to-peak, max cycle range, or pseudo-damage increase — enforcing a physics prior that a purely data-driven fit wouldn't otherwise capture. Random Forest won, with CV MAPE = 0.100 ± 0.024 , ahead of ElasticNet (0.127), Ridge (0.170), SVR (0.172), XGBoost (0.176), HistGradientBoosting (0.185), and KNN+PCA (0.263). The reported metric is the organisers' own $\max(0, 1 - \text{MAPE})$ score (Info Kit Section 5): 0.900 under nested CV. This result is disclosed honestly rather than oversold: train-set MAPE (0.033) is

substantially lower than the nested CV estimate (0.100), an overfit-gap warning the pipeline surfaces automatically rather than hides. It is the nested CV figure — not the more flattering in-sample one — that is reported as the honest estimate of performance on unseen data.

Flagging rather than fabricating: unlike Door, Rail, and ACV, the source material doesn't frame SHM's validation choice as an explicit "brief left this open" design decision — the RepeatedKFold structure is stated as method, with no accompanying assumption rationale. If you want a matching Assumptions paragraph here, let me know what open decision it should cover (e.g. why RepeatedKFold over a grouped split, given each file is presumably one independent unit as in Rail) and I'll draft it rather than invent a rationale not in your source.

[3] App Building

RailGuard

Persona Mapping & Model Application Report

Predictive Subsystem AI & Anomaly Detection — Tuas West Depot

1. App Overview

RailGuard follows a consistent 3-step workflow across all four subsystems (ACV, Door, Rail Corrugation, SHM): Select Subsystem → Upload Data File → Subsystem Diagnostics. Every screen shares the same shell — depot selector, user identity (Alex Tan, Lead Depot Technician), an Auto/Desktop/Mobile view toggle, an "Analysis Complete • High Confidence" status badge, a CENELEC EN 50126 compliance mark, and a "Download Prediction & Diagnostic Report (PDF)" action.

2. Persona → App Feature Mapping

Persona / Pain Point	Feature Named in Persona Doc	What's Actually Visible in the App
All stakeholders Fragmented data, hard to prioritize, tight maintenance windows	Colour-coded ranking, train schematic diagram, drill-down hierarchy	Direct match: the Car 1–8 carriage grid (ACV screen) colour-codes each car green (OK) / red (FAULT), with a "Car 3 Needs Fix" summary badge — lets a user zoom straight to the flagged car without reading raw telemetry. Consolidating all four subsystems behind one Select-Subsystem tab also directly answers the "many substations, high data volume" fragmentation complaint.
Tan Boon Huat Senior Technician — alert trust, disconnected systems, tacit knowledge	Alert Confidence Ledger, Unit Memory Card, Action-Tier Translator + Evidence Snapshot	Partial match: "Analysis Complete • High Confidence" is a confidence signal, but it's a single global badge, not a per-unit accuracy history with confirm/false-alarm feedback as specified for the Alert Confidence Ledger. The diagnostic cards (e.g. "Car 3 compressor pulling 18.6A... exceeds 12.0A continuous ceiling") do function as an Action-Tier Translator — plain-language reasoning with the actual reading and threshold. No Unit Memory Card (historical timeline of past faults/repairs) is visible

Persona / Pain Point	Feature Named in Persona Doc	What's Actually Visible in the App
		in any of the four screens.
Wei Jie Tan Junior Technician — escalation anxiety, 4–5 disconnected systems, poor signal	Evidence Snapshot, Unified mobile app, Action-Tier Translator	Partial match: the Desktop/Mobile toggle directly answers the "unified app instead of 5 systems" pain point. The Door screen's "What to do right now" + numbered "Recommendations & Action Guidance" (Isolate & Clear Rail Track → Perform Precision Inspection → Verify Laser Tolerance & Re-test) is a strong Action-Tier Translator implementation. The PDF report could serve as an Evidence Snapshot, but the persona doc calls for a one-tap, chart+confidence+precedent summary — the PDF is a fuller report, not confirmed as that lighter-weight artifact.
Lim Siew Hoon OCC Duty Manager — alert fatigue, fleet-wide framing, escalation calls	Fleet Risk Triage, Escalate & Reach + Audit log, Generate report	Partial match: "Download Prediction & Diagnostic Report (PDF)" is a direct match for "Generate report." The Car 1–8 grid is risk-triage-shaped but only within one train, not sorted by urgency across the fleet as the persona doc describes. No Escalate & Reach button or audit log is visible in any screen shown.

3. Model Application — How Each Trained Model Drives Its Screen

Subsystem	Model (from approach write-up)	What the Screen Shows	Mapping
ACV	Mahalanobis distance, per-car anomaly ranking (nested LOFO-CV, rank-decay score 0.979)	Car 3 flagged red among 8 cars; three root-cause cards (Compressor Current, Evaporator Delta-T, Refrigerant Pressure)	The car-grid colouring is the ranking model's direct output — Car 3 is the top-ranked anomalous car. See flag in Section 4 on the root-cause cards.

Subsystem	Model (from approach write-up)	What the Screen Shows	Mapping
Door	L1 logistic regression, Normal/Abnormal classification + gap-based segmentation	"Abnormal Resistance" banner; three feature cards (Closing Push Force, Closing Time, Motor Power Draw) each with a reading, threshold, and % deviation	Matches directly — the three cards correspond to summary-statistic features (force, timing, current draw) that feed the classifier, surfaced as human-readable explanations rather than raw model coefficients.
Rail Corrugation	SVM (RBF), 3-class Normal / Side I / Side II (macro F1 0.621)	Side I panel: green "Normal Profile"; Side II panel: red "Corrugation Affected, FAULT · GRIND REQUIRED"	Matches directly — Side I and Side II are judged independently from the same file, exactly as shown in the two-panel layout.
SHM	Random Forest regression, cumulative fatigue-damage estimate (MAPE-based score 0.900)	"68% WORN" gauge; "Bogie Condition Safe," next check at 750,000 km	Matches directly — a regression output (continuous damage estimate) displayed as a percentage gauge, consistent with SHM being framed as regression rather than classification/ranking.

4. Flag Worth Resolving Before Presenting

The ACV screen's root-cause cards display Compressor Current and Refrigerant Pressure as explanatory evidence for the flagged car. But the ACV write-up explicitly states these two feature families (refrigerant-pressure and compressor-cycling) were deliberately excluded from the ranking model after identifying circular-validation risk — only one training file had that richer schema, so there was no independent file to test whether those features generalise.

This isn't necessarily a contradiction — the UI could legitimately be pulling these as raw supplementary telemetry readings for human context, separate from what the ranking model itself used (the 8 robust-z-scored features: cooling-gap mean/p90/p99, fault rate, mode/info-invalid rates, load-active rate, missing-data rate). But as currently displayed, a reader (or a judge reviewing both the write-up and

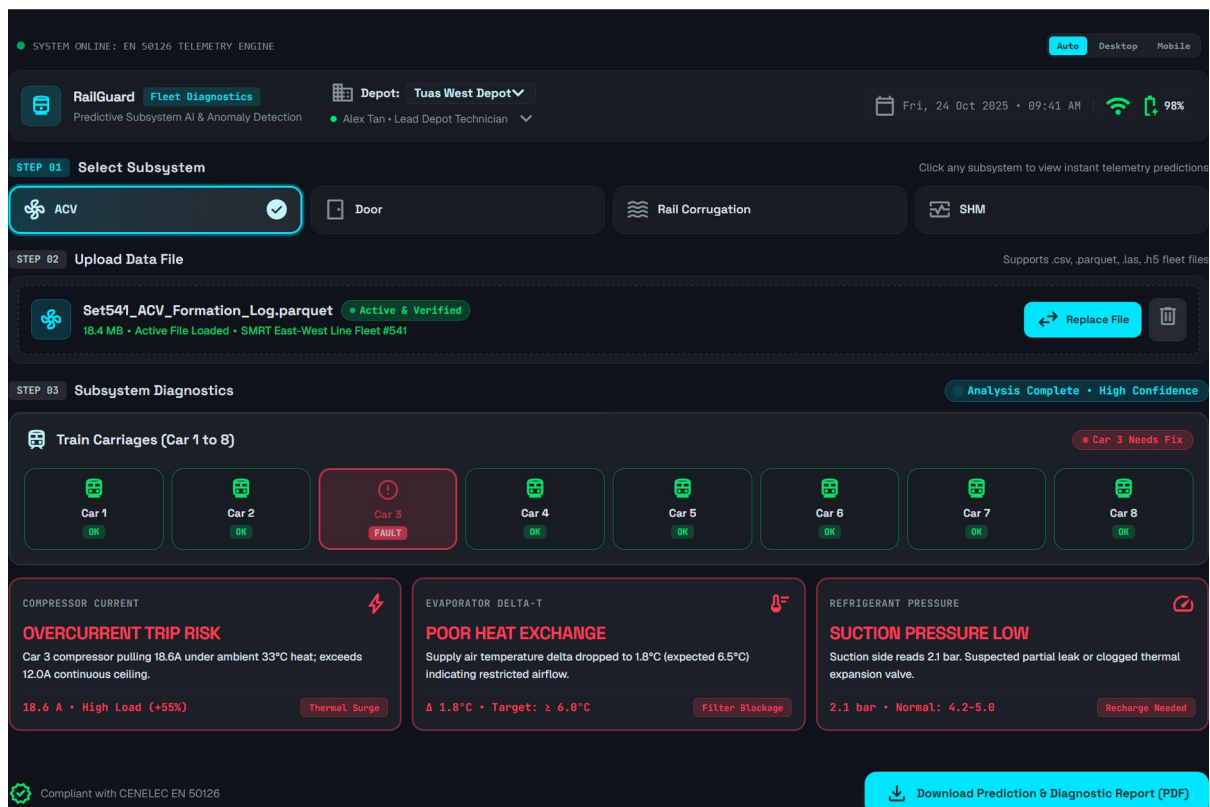
the demo) could reasonably conclude the pressure/compressor features were used in the model, directly undercutting the "leakage-audited feature set" disclosure — one of the strongest technical-honesty points in the write-up. Worth either relabelling those cards as "supplementary telemetry" distinct from "model features," or confirming they are genuinely excluded from the scoring model and only shown post-hoc.

5. What's Not Confirmed From These Four Screens

To avoid overstating the mapping: the Alert Confidence Ledger's accuracy history and feedback loop, the Unit Memory Card, the human-flag-AI-error review feature, Escalate & Reach, the audit log, and fleet-wide risk triage across multiple trains are all named in the persona work but not visible in any of the four screens shown here. If these exist on other screens (e.g. an alerts inbox, a unit detail view, or the Mobile mode specifically), they should be folded into this mapping — otherwise they currently read as roadmap items rather than shipped features.

Appendix — App Screens Referenced

ACV — Refrigerant-Leak Car Localisation



Car 3 flagged (red) among 8 cars, with compressor current, evaporator delta-T, and refrigerant pressure diagnostic cards.

SYSTEM ONLINE: EN 50126 TELEMETRY ENGINE

AutoDesktopMobile

RailGuardFleet DiagnosticsPredictive Subsystem AI & Anomaly Detection

Depot: Tuas West DepotAlex Tan • Lead Depot Technician

Fri, 24 Oct 2025 • 09:41 AM98%

STEP 01Select Subsystem

Click any subsystem to view instant telemetry predictions

ACVDoorRail CorrugationSHM

STEP 02Upload Data File

Supports .csv, .parquet, .las, .h5 fleet files

Set541_Door4_Telemetry.csvActive & Verified42.8 MB • Active File Loaded • SMRT East-West Line Fleet #541Replace File

STEP 03Subsystem Diagnostics

Analysis CompleteHigh Confidence

Abnormal ResistanceDoor subsystem Abnormal Resistance detected. Object or dirt in track.ACTION NEEDED

CLOSING PUSH FORCEPUSHING TOO HARDMotor forcing against blockage with 340 N thrust, nearly double safe limit.
340 N • Exceeds threshold (+8%)Excessive Resistance

CLOSING TIMECLOSING TOO SLOWLYDoor took 4.8s to close — 50% slower than nominal cycle due to track friction.
4.8 sec • Max Limit: 3.2s (+50%)Abnormal Cycle Duration

MOTOR POWER DRAWMOTOR OVERSTRAININGHeavy electrical stress detected: drawing 14.2A compared to the safe 7.0A ceiling.
14.2 A • Normal range: 4.5–7.0 ACurrent Spike Detected

What to do right now:
Check guide rails & clean debris around door drive belt.Inspect Rail Now

RECOMMENDATIONS & ACTION GUIDANCEURGENT RECTIFICATION

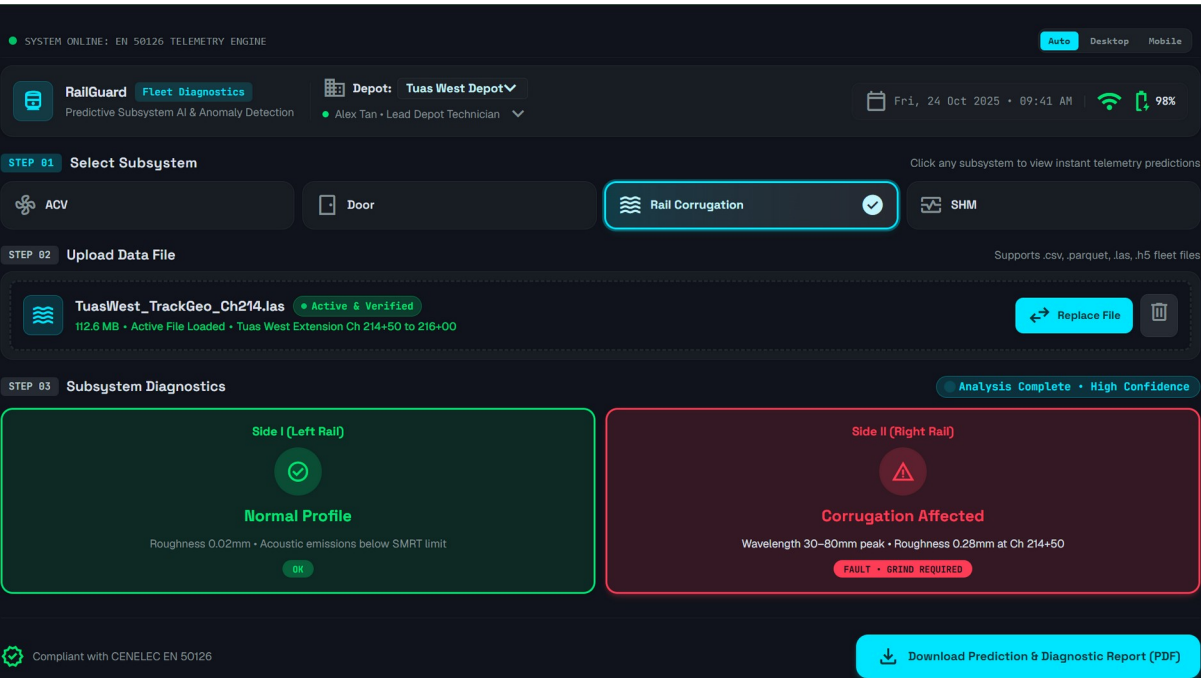
1 Isolate & Clear Rail Track
Check guide rails & door runner track for obstruction or foreign debris.

2 Perform Precision Inspection
Inspect drive belt tension and motor actuator linkage before manual cycling.

3 Verify Laser Tolerance & Re-test
Perform optical clearance calibration & automated 5-cycle door stroke test.

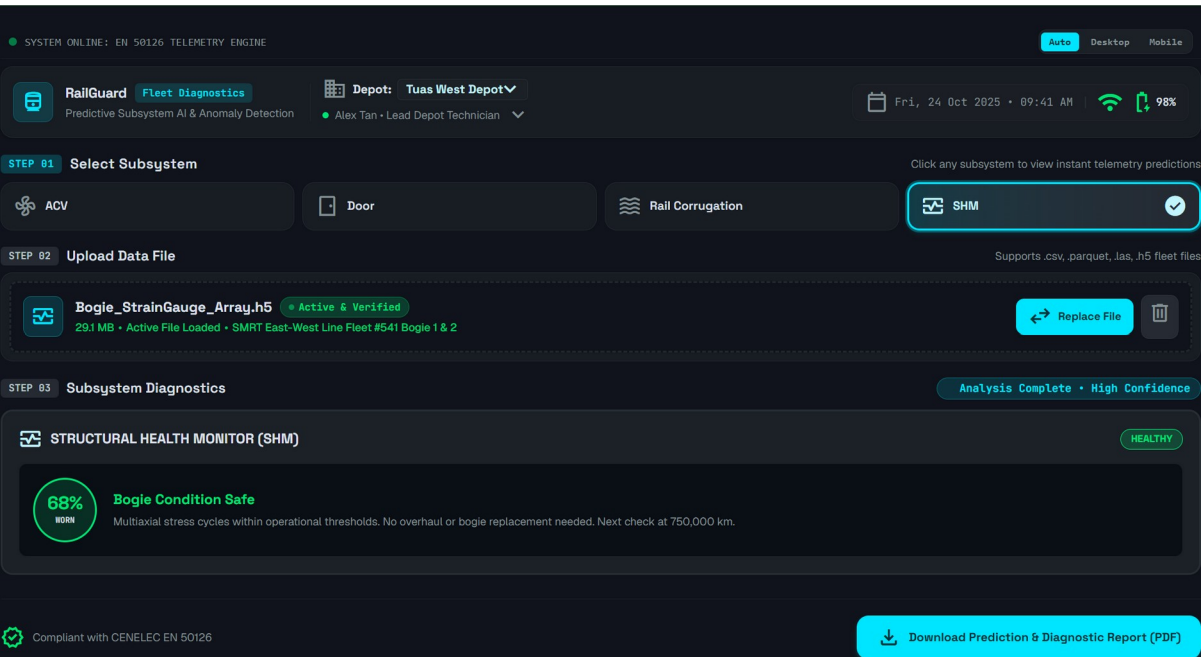
Compliant with CENELEC EN 50126Download Prediction & Diagnostic Report (PDF)

Rail Corrugation — Side I / Side II Classification



Side I (Left Rail) Normal Profile vs. Side II (Right Rail) Corrugation Affected — FAULT, grind required.

SHM — Bogie Fatigue-Damage Estimate



68% worn gauge with "Bogie Condition Safe" status and next-check mileage.